

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457832.9088 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.015
Transit depth (Rp/Rs)^2	2.91 +/- 0.51 [%]
Orbital Inclination (inc)	87.75 +/- 0.87
Airmass coefficient 1 (a1)	1.142 +/- 0.022
Airmass coefficient 2 (a2)	-0.094 +/- 0.02
Scatter in the residuals of the lightcurve fit is	1.95 %
Best Comparison Star	#2 - [456, 132]
Optimal Aperture	3.21
Optimal Annulus	12.71
Transit Duration (day)	0.1207 +/- 0.0092

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.171 ± 0.015 vs 0.1457 (deviation 1.44σ)
Duration fit vs published	0.1207 d vs 0.12 d (deviation 0.06σ)
Mid-time O–C	-4.1 min
Depth SNR	9.7
Residual scatter	1.95 %

Light curve

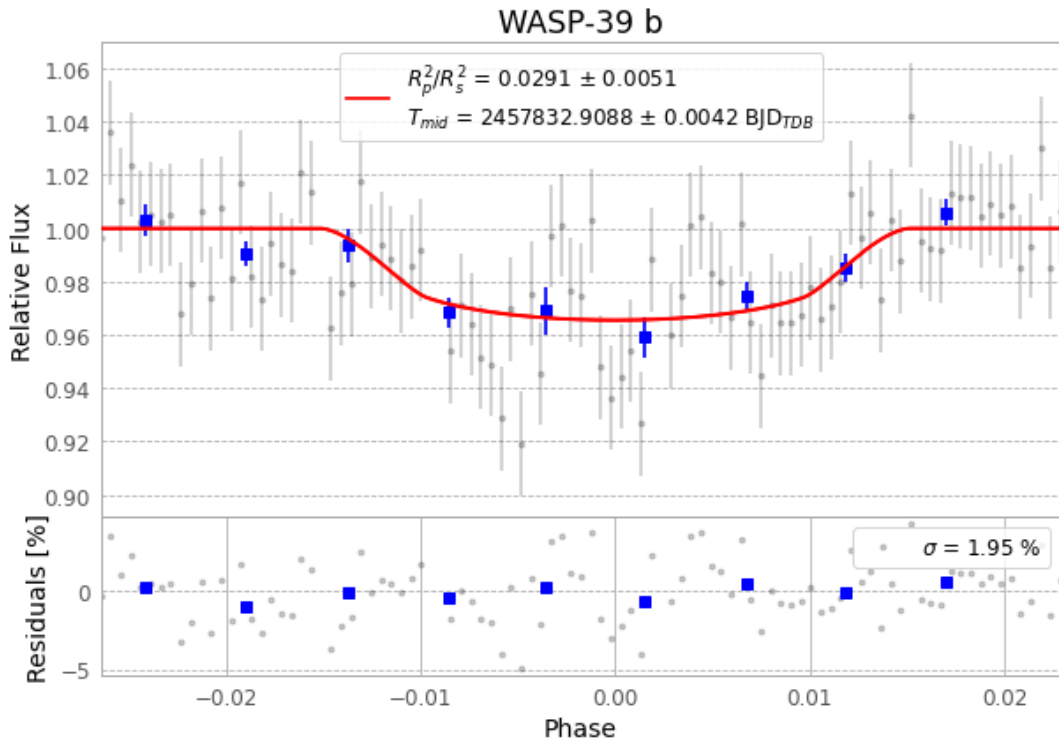


Figure 1 — FinalLightCurve_WASP-39 b_2017-03-20.png (SHA-256 d8542004a93d085b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [150, 227], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 45a399353deb0fe7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

96 FITS frames (63 MB), WASP-39170320070612.FITS → WASP-39170320115412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-170320.log (345993df0602...), FinalLightCurve_WASP-39 b_2017-03-20.png (d8542004a93d...), FinalLightCurve_WASP-39 b_2017-03-20.csv (bb7c39d84e34...)

Compute resources

Wall time 190.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 21:58Z.